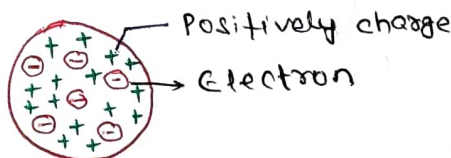


ATOM

J.J Thomson's Atomic Model

- Atom is spherical, electrically neutral [+ve charge = -ve charge]
- +ve charge of atom also uniformly distributed in which electrons are embedded in between.
- Mass of atom uniformly distributed inside volume of atom.



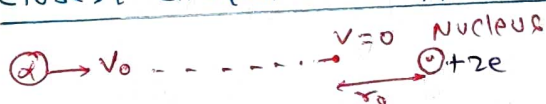
* Drawbacks of Thomson's Model

- Can't explain → ① Spectrum of atoms
② scattering of α -particles

Rutherford's Atomic Model

- α -particle scattering experiment.
- Most α -particle went straight through i.e. atom is hollow.
- 1 out of 8000 deflected by 180°
- Entire +ve charge & almost whole mass of atom concentrated at centre → Nucleus.
- The electrons could not deflect the path of a α -particle i.e. electrons are very light.

closest distance of approach



COME

$$(K.E + P.E)_i = (K.E + P.E)_f$$

$$\frac{1}{2} M_\alpha v^2 + 0 = \frac{K z e (2e)}{r_0} + 0$$

$$r_0 = \frac{4 K z e^2}{M_\alpha v^2}$$

$$\text{Radius of atom } (r_{\text{atom}}) = 10^{-10} \text{ m} = 1 \text{ \AA}$$

$$\text{Radius of Nucleus} = 10^{-15} \text{ m} = 1 \text{ fermi}$$

* Drawback of Rutherford's Model

- Can't explain → ① Line Spectrum of Atom
② Stability of Atom.

Bohr's Atomic Model

① 1st Postulate

- e^- in atom could revolve in certain stable orbit without emission of radiant energy.
- Each possible state has definite total energy i.e. stationary states of atom.

② 2nd Postulate

- The electron revolves around the nucleus only in those orbits for which

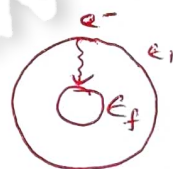
$$L = \frac{n h}{2 \pi}$$

$L \rightarrow$ Angular momentum
 $h \rightarrow$ Planck's constant

$$n \rightarrow 0, 1, 2, \dots$$

③ Bohr's 3rd postulate

- e^- might make a transition from one of its specified non-radiating orbits to another of lower energy.
- When it does so, a photon is emitted having energy equal to the energy diff. b/w the initial & final states.



$$E_i - E_f = E_{\text{photon}}$$

Formula Analysis:-

Velocity of e^- in n th orbit

$$V = \left(\frac{e^2}{2 \epsilon_0 h} \right) \frac{Z}{n} = (2.18 \times 10^6) \frac{Z}{n} \text{ m/s}$$

$$V = \frac{c}{137} \times \frac{Z}{n}$$

$$\begin{aligned} V &\propto Z \\ V &\propto \frac{1}{n} \end{aligned}$$

Radius of the n th orbit

$$r = 0.53 \times \frac{n^2}{Z} \text{ \AA}$$

$$\begin{aligned} r &\propto n^2 \\ r &\propto \frac{1}{Z} \end{aligned}$$

Time period of revolution of e^- in n th orbit

$$T = \frac{2 \pi r}{V}$$

$$T \propto n^3$$

$$T \propto \frac{1}{Z^2}$$

$$T = 1.51 \times 10^{-16} \frac{n^3}{Z^2}$$

Frequency of revn of e^- in n th orbit

$$f = \frac{2}{3} \times 10^{16} \frac{Z^2}{n^3}$$

$$f = \frac{1}{T} \propto \frac{Z^2}{n^3}$$

Magnetic field at centre of nth orbit

$$B \propto \frac{z^3}{n^5}$$

Total Energy of e^- in nth orbit

$$KE = \frac{ze^2}{8\pi\epsilon_0 r}$$

$$PE = \frac{-ze^2}{4\pi\epsilon_0 r}$$

$$TE = KE + PE$$

$$TE = -KE = \frac{PE}{2}$$

$$PE = -2KE = 2TE$$

for H-atom

$$TE = -13.6 \text{ eV} \frac{z^2}{n^2}$$

$$TE = \frac{-13.6 \text{ eV}}{n^2}$$

Energy of Each level of H-atom

n	State	Orbit		Energy
1	Ground state	1st orbit	K	-13.6 eV
2	1st excited state	2nd orbit	L	-3.4 eV
3	2nd excited state	3rd orbit	M	-1.51 eV
4	3rd excited state	4th orbit	N	-0.85 eV
5	4th excited state	5th orbit	O	-0.54 eV

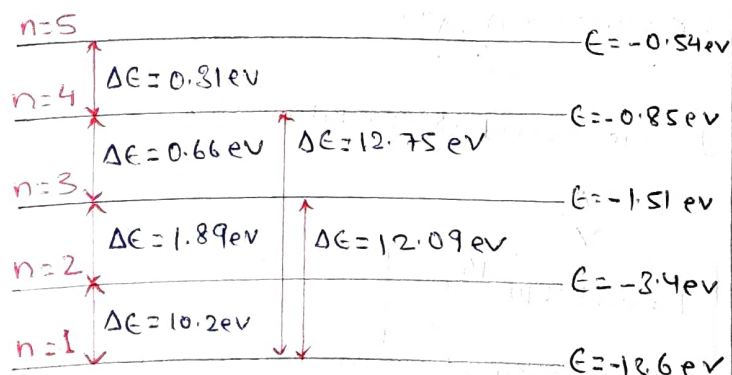
H-Spectrum (Emission Spectrum)

$$\Delta E_{\text{loss}} = E_i - E_f$$

$$E_{\text{photon}} = \frac{me^4}{8\epsilon_0^2 h^2} \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$\times \frac{1}{\lambda} \rightarrow R$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$



SL No.	Spectral Series	Region	n_f	n_i
1	Lyman	U.V ray	1	2, 3, 4, ...
2	Balmer	Visible	2	3, 4, 5, ...
3	Paschen	Intra-red	3	4, 5, 6, ...
4	Brackett	I.R	4	5, 6, 7, ...
5	P-fund	I.R	5	6, 7, 8, ...
6	Humphrey	far I.R	6	7, 8, 9, ...

1st line

α -line

λ_{max}

f_{min}

E_{min}

Last line

$n_i = \infty$

λ_{min}

f_{max}

E_{max}

* No. of Spectral line

$$N = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

$$N = \frac{n(n-1)}{2}$$

Ionization Energy -

Min. energy required to remove an e^- from isolated atom.

$$I.E = |TE|$$

* Drawbacks of Bohr's Model

- Couldn't explain fine structure of spectral lines.
- Splitting of spectral lines in presence of magnetic field (Zeeman effect) & electric field (Stark effect) which is not explained.
- Only valid for single e^- atom (H, He^+, Li^{2+})
- Not explain the intensity of spectral lines.
- Only considered particle nature of e^- .

$$\Rightarrow \text{wave number} = \frac{1}{\lambda}$$

$$\Rightarrow \text{Angular wave no.} = \frac{2\pi}{\lambda} = K$$

Atomic collision

$$\Delta K.E = \left(\frac{4n}{1+n} \right)^2$$

$$n = \frac{M_{atom}}{M_{collide}}$$

$$(\Delta K.E)_{max} = 10.02 \text{ eV} = 12.09 \text{ eV}$$

→ Collision is Inelastic
otherwise collision is Elastic

X-RAY

→ Inverse process of photoelectric effect.
→ collection of photon.

→ Energy of photon = 0 to $K.E_{max}$

$$E_{max} = \frac{hc}{\lambda_{min}} = K.E_{max} e^-$$

$$\frac{hc}{\lambda_{min}} = eV_0 \rightarrow \lambda_{min} = \frac{hc}{eV_0}$$

accelerating voltage

→ cutoff wavelength

Intensity of X-ray $\propto$ tube current
frequency of X-ray $\propto$ tube voltage

X-ray :- (Two types)

(i) Soft X-ray ($\lambda = 4 \text{ \AA}$)

→ low penetrating power at low potential difference

(ii) Hard X-ray ($\lambda = 1 \text{ \AA}$)

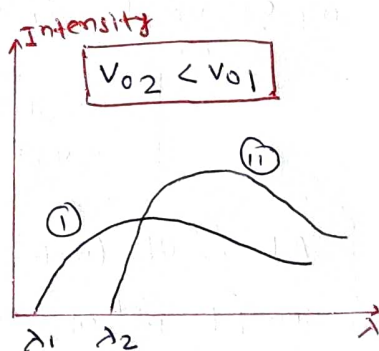
→ High penetrating power at high potential.

X-ray spectrum

(a) Continuous X-ray spectrum

→ due to retardation of e^-

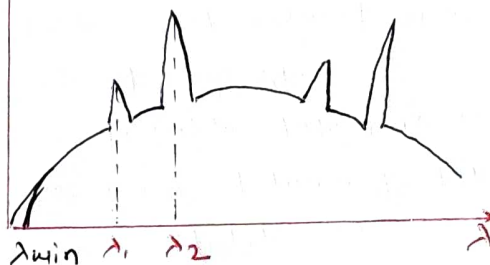
$$\lambda_{min} \propto \frac{1}{V_0}$$



(b) characteristic X-ray spectrum

I

$z = \text{high}$



$$\lambda \propto \frac{1}{(z-1)}$$

$$\sqrt{f} \propto (z-1)$$



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